Walks, paths and connectivity from the adjacency matrix of a graph

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Abstract

This study note comes from a graph theory course. A lecture whiteboard of that course poses two questions about the adjacency matrix A of a graph with N vertices, a table of zeros and ones that has a 1 where two vertices are joined by an edge, and the assignment adds a third. First, the entry in row i and column j of the matrix power A^k is the number of walks of length exactly k from vertex n_i to vertex n_j; it counts walks, not paths. Second, a graph is connected exactly when every off-diagonal entry of $A + A^2 + \cdots + A^{N-1}$ is positive, which is the same as saying that $(I + A)^{N-1}$ has no zero entry. Third, a path between n_i and n_j, in the convention of the course, is a walk from n_i to n_j in which no vertex occurs twice. Each result is proved here, using the course textbook's definitions and numbering, and is shown on a connected and on a disconnected example with three vertices. The board's drawing and its matrix differ by one edge; the note follows the matrix.

1 Introduction

A graph is a set of points, called vertices, together with lines, called edges, that join some pairs of points. A computer often stores a graph as a table of zeros and ones, the adjacency matrix A, which has a 1 in row i and column j when vertices n_i and n_j are joined by an edge. This note asks what can be read from that table alone. It has three questions.

(Q1) What does the entry $(A^k)_{ij}$ of the k-th power of A represent?

(Q2) How can one decide, using only A, whether the graph is connected?

(Q3) What is the definition of a path between the vertices n_i and n_j?

Where the questions come from. Questions (Q1) and (Q2) are reconstructed from a lecture whiteboard, photographed twice from slightly different angles [4, 5]. The board reads, in the handwriting's own words, "What does A_{ij} represent" for question 1, with a small mark above the letter A, and "How to know if graph is connected or not" for question 2. The same board gives the definition "Graph is called connected if for any n_i, n_j there exists a path between nodes" [4]. The small mark is read as an exponent k, so that question 1 becomes (Q1), because the assignment asks about A^k (Convention 3 below); this reading is an interpretation. Question (Q3) is not on the board. It comes from the assignment text, as reported to the author of this note; that text is not reproduced here, so (Q3) is best taken as the third question this note has chosen to answer.

Why it matters. A table of zeros and ones looks like it holds only direct connections. Question (Q1) shows that multiplying the table by itself answers a richer question, namely how many ways there are to travel between two vertices in a fixed number of moves. Question (Q2) turns that into a yes or no answer about the whole graph, and question (Q3) fixes the word "path" on which the board's definition of connected rests.

Short answers. Each is stated with its proof later.

(A1) For every integer $k \geq 1$, the entry $(A^k)_{ij}$ counts the walks of length exactly k from n_i to n_j (Theorem 4.1). It counts walks, not paths (Theorem 4.4).

(A2) A graph with N vertices is connected if and only if every entry of $A + A^2 + \cdots + A^{N-1}$ that is not on the diagonal is positive (Theorem 5.1). Equivalently, $(I + A)^{N-1}$ has no zero entry (Theorem 5.3), where I is the identity matrix.

(A3) In the convention of the course, a path between n_i and n_j is a sequence of vertices that starts at n_i, ends at n_j, has an edge between each two neighbours in the sequence, and never repeats a vertex (Definition 2.34).

The story used throughout. Every definition, theorem, test, and example is preceded by a paragraph headed "In plain words". These paragraphs tell one story: a country of towns joined by two-way roads. The paragraphs use only the story words of Table 1, one story word for each technical term. The letter N is the number of towns, and k is a number of steps.

Table 1: Dictionary between the technical terms of this note and the story words used in the plain-words paragraphs.

Technical term	Story word	Meaning in the story
vertex	town	a place
edge	road	a two-way road between two different towns, at most one per pair
adjacent	joined	two towns with a road between them
adjacency matrix	road table	row i, column j holds 1 if towns i and j are joined, else 0
degree	number of roads	the number of roads that leave one town
walk	trip	a journey road by road, turning back allowed
length	number of steps	one step is one road driven
closed walk	round trip	a trip that ends where it began
trail	no-repeat-road trip	a trip that never drives the same road twice
path	no-revisit trip	a trip that never visits a town twice
cycle	clean round trip	a round trip, at least one step long, that visits no town twice except the start, which is also the end
distance	fewest steps	the smallest number of steps of any trip between two towns
connected	one group	every town can reach every other town
component	group of towns	towns that can reach each other, and no others

Conventions and assumptions. The following items fix the readings and choices of this note. Later text refers to them by number.

1. *Graphs.* Every graph is finite, simple, and undirected. Simple means no edge joins a vertex to itself and no two edges join the same pair. The assignment text states this, and it fits the board matrix, which is symmetric with zeros on its diagonal.

2. *Notation.* The board writes the graph as (G, V) and an edge as $e_{ij} \in V$. The course textbook writes $G = (V, E)$ with E the set of edges [8, Def. 1.1.1, p. 9]. This is a difference of notation only, and this note uses the textbook's $G = (V, E)$. Vertices are named $n_1, \ldots, n_N$ as on the board, where N is the number of vertices; the textbook names them v_i [8, p. 11].

3. *The mark above A.* The mark is read as an exponent k. This is an interpretation: the photographs show only a small mark, and the assignment text asks about A^k. If the mark is instead a quotation mark, question 1 asks what A_{ij} represents; that is answered by Definition 2.9 and by the case $k = 1$ of Theorem 4.1.

4. *The example graph.* The board shows a drawing and a 3×3 matrix that do not describe the same graph, as Table 3 in Section 6 sets out edge by edge. The matrix is taken as the example, because the problem statement on the board says that the graph is given through its adjacency matrix. The drawing as read is shown in Figure 2. The arrowhead on the line from 1 to 2 is read as a stray mark, because the matrix is symmetric and the problem concerns undirected graphs. The note does not claim to know what the lecturer intended.

5. *The word "path".* In the course, a path has no repeated vertex. This is the textbook's simple path, Definition 1.2.13 [8, p. 22]. The textbook's own word "path" (Definition 1.2.12) means a walk with no repeated edge; this note calls that a *trail*, with the textbook's number beside it where the textbook's wording is meant. In short: the textbook's "path" is this note's trail, and the textbook's "simple path" is this note's path. Table 2 compares the conventions.

6. *Length zero.* A single vertex counts as a walk of length 0. This is the note's own convention. The textbook's Definitions 1.2.10 to 1.2.16 do not say whether length 0 is allowed [8]. Two of its statements assume a trail of length 0 from a vertex to itself, namely that $d(u, v) = 0$ if and only if $u = v$ (p. 23) and that its relation $\sim$ is reflexive (p. 49) [8, pp. 23, 49], so this convention agrees with the textbook's usage. Grinberg also allows length 0 [3, Def. 2.9.1(a), p. 36]. A cycle always has positive length. Without this convention, "any two vertices" in the definition of connected would have to be read as two different vertices (Remark 5.6).

7. *Page numbers.* Pages of the textbook are printed page numbers, which are the PDF page numbers minus one. The textbook is in Armenian; the definition numbers are the book's own. Its term for the adjacency matrix is, in transliteration, *harevanutyan matrits*; the Armenian script is not typeset in this note. A reader who cannot read Armenian cannot check the textbook citations directly, so every definition taken from the book is restated in English here and its number and page are given.

8. *Kinds of statement.* A definition carries the number of the source it comes from. A lemma or theorem followed by a proof is proved here; the sources that also state it are cited in its first sentence. The textbook contains no walk-counting theorem and no matrix test for connectivity. A text search of its 216 pages for the term for adjacency matrix finds the term only on printed pages 7, 11, 12, 34, and 61, and the text layer of these pages contains neither result [8]. This is the result of a search, not a proof of absence, so both results are taken from other sources and proved here.

Organization. Section 2 introduces the definitions in the textbook's order. Section 3 answers question (Q3) by comparing conventions and proves that every walk contains a path. Section 4 answers question (Q1). Section 5 answers question (Q2), and Section 6 works a connected and a disconnected example and reports the board mismatch. Section 7 gives limits and a comparison with the sources, and Section 8 gives the answers. Appendix A lists the sources, the pages checked, and how to rerun the calculations.

2 Definitions

Notation. Throughout, $G = (V, E)$ is a finite simple undirected graph (Convention 1) with vertex set $V = \{n_1, \ldots, n_N\}$ and $N \geq 1$. The letters u and v stand for any vertices. The letters i, j, r index vertices, p, q, s, t are positions in a list of vertices, and k is a length. The definitions follow the order of the textbook, and each carries the textbook's number when it has one. After each definition come a remark on what it allows and forbids, an example, and a non-example. Later proofs use the definitions of graph, adjacency matrix, matrix power, degree, walk, trail, path, and connected; the definitions of closed walk, cycle, and distance are given because the textbook gives them in this order and because the comparison of naming conventions in Section 3 needs them (a cycle is also used in one remark). The running example is the *board graph* of Figure 1: three vertices and the two edges n_1n_2 and n_2n_3 (Convention 4).

$$n_1 \text{ ——— } n_2 \text{ ——— } n_3$$

Figure 1: The board graph: vertices n_1, n_2, n_3 and edges n_1n_2 and n_2n_3, the graph described by the matrix on the board.

In plain words. A country has towns and roads. Each road joins two different towns and runs both ways. No two roads join the same pair of towns. No road starts and ends in the same town.

Definition 2.1 (graph; textbook Definition 1.1.1)**.** A *graph* $G = (V, E)$ consists of a finite set V of *vertices* and a set E of *edges*, each edge being an unordered pair of two different vertices [8, Def. 1.1.1, p. 9]. This is the definition of a simple undirected graph [6, Def. 1.1.1, p. 2]. The edge joining n_i and n_j is written n_in_j; the board writes it e_{ij}.

Remark 2.2. Because E is a set of unordered pairs of different vertices, there are no loops and no repeated edges, and the edge n_in_j is the same as the edge n_jn_i.

Example 2.3. The board graph of Figure 1 has $V = \{n_1, n_2, n_3\}$ and $E = \{n_1n_2,\ n_2n_3\}$.

Non-example 2.4. Two roads between the same two towns, or a road from a town back to itself, do not give a simple graph.

In plain words. Two towns are joined when a road runs between them. Towns with no road between them are not joined, even if a third town links them.

Definition 2.5 (adjacent; textbook Definition 1.1.4)**.** Two vertices n_i and n_j are *adjacent* if n_in_j is an edge [8, Def. 1.1.4, p. 11].

Remark 2.6. Adjacency concerns one edge only. A vertex is never adjacent to itself in a simple graph.

Example 2.7. In the board graph, n_1 and n_2 are adjacent.

Non-example 2.8. In the board graph, n_1 and n_3 are not adjacent, although a trip from n_1 to n_3 exists through n_2.

In plain words. The road table has one row and one column for each town. Look at the row of one town and the column of another. Write 1 if a road joins them and 0 if not. The table looks the same if rows and columns are swapped, because roads run both ways. Every cell from a town to itself holds 0.

Definition 2.9 (adjacency matrix; textbook, after Definition 1.1.6)**.** The *adjacency matrix* of G is the $N \times N$ matrix A with $A_{ij} = 1$ if n_i and n_j are adjacent and $A_{ij} = 0$ otherwise [8, p. 11]. The board writes this rule with e_{ij} in place of adjacency. The textbook gives the matrix no number of its own and never writes e_{ij}.

Remark 2.10. Every entry is 0 or 1. The diagonal is zero, $A_{ii} = 0$, and the matrix is symmetric, $A_{ij} = A_{ji}$ [8, p. 11]. The first holds because there are no loops, the second because an edge is unordered.

Example 2.11. The board graph has

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

which is the matrix written on the board.

Non-example 2.12. The matrix with rows $(0,1,0)$, $(0,0,1)$, $(0,0,0)$ is not symmetric, so it is not the adjacency matrix of any simple undirected graph. A matrix with a 1 on its diagonal is not one either.

In plain words. To multiply two tables, take one row of the first and one column of the second. Multiply their cells in pairs, going along together, and add the results. That sum is the new cell. Doing this twice gives the table for two steps, three times gives three steps, and so on.

Definition 2.13 (matrix product and power)**.** For $N \times N$ matrices X and Y, the product XY has entry $(XY)_{ij} = \sum_{r=1}^{N} X_{ir} Y_{rj}$, which combines row i of X with column j of Y. The powers of A are $A^0 = I$, the identity matrix with ones on the diagonal and zeros elsewhere, and $A^{k+1} = A^k A$ for $k \geq 0$.

Remark 2.14. The sum runs over every vertex n_r. It adds the products of two entries, one from each factor, for each r.

Example 2.15. For the board matrix, $(A^2)_{11} = A_{11}A_{11} + A_{12}A_{21} + A_{13}A_{31} = 0 + 1 + 0 = 1$.

Non-example 2.16. The matrix A^2 is not the matrix whose entries are the squares of the entries of A. Squaring each entry of the board matrix gives A again, while $(A^2)_{22} = 2$ (see Section 6).

In plain words. The number of roads at a town is the number of ways to leave it by one road. The road table shows this number as the sum of one row.

Definition 2.17 (degree; textbook Definition 1.2.1)**.** The *degree* $\deg(n_i)$ of a vertex is the number of edges at it [8, Def. 1.2.1, p. 16]. The textbook writes $d(v)$; this note writes deg so that the letter d is kept for distance (Definition 2.42).

Remark 2.18. A vertex of degree 0 is called isolated here; it is adjacent to nothing.

Example 2.19. In the board graph, $\deg(n_1) = 1$, $\deg(n_2) = 2$, and $\deg(n_3) = 1$.

Non-example 2.20. The number of vertices N is not a degree. The board graph has $N = 3$ and no vertex of degree 3.

In plain words. Add up one row of the road table. You get the number of roads at that town. Add up all the rows. You count every road twice, once from each end.

Theorem 2.21 (row sums are degrees)**.** *For every i, $\sum_{j=1}^{N} A_{ij} = \deg(n_i)$. Consequently the sum of all entries of A is $2|E|$, twice the number of edges. The second statement is the textbook's Theorem 1.2.1, that the sum of all degrees is $2|E|$ [8, p. 16], which the textbook proves on p. 16 and which is not proved again here. The first statement follows from the textbook's rule for A [8, p. 11]; its proof is given here. (The textbook's Definition 1.2.1 is the definition of degree, and its Theorem 1.2.1 is this sum.)*

Proof. Fix i. By Definition 2.9, $A_{ij} = 1$ exactly when n_j is adjacent to n_i. So $\sum_j A_{ij}$ is the number of neighbours of n_i. Send each edge at n_i to its other end. No edge at n_i returns to n_i because there are no loops, and two different edges at n_i never reach the same neighbour because there are no repeated edges. So this assignment is a one-to-one correspondence between the edges at n_i and the neighbours of n_i, and the number of neighbours is $\deg(n_i)$. Adding over i gives $\sum_i \deg(n_i)$, which is $2|E|$ by the textbook's Theorem 1.2.1 (the sum of degrees). □

In plain words. A trip goes from town to town along roads. You may turn round and drive a road again. You may pass through a town as often as you like. The number of steps is the number of roads driven, and a road driven twice counts twice.

Definition 2.22 (walk; textbook Definition 1.2.10)**.** A *walk of length k from u_0 to u_k* is a sequence $u_0, e_1, u_1, \ldots, e_k, u_k$ of vertices and edges with $e_s = u_{s-1}u_s$ for $1 \le s \le k$ [8, Def. 1.2.10, p. 21]. In a simple graph a walk is determined by its vertices, and the textbook also writes a walk as the list $u_0, u_1, \ldots, u_k$ [8, p. 21]. The *length* is the number of edges, counted each time an edge occurs.

Remark 2.23. Vertices and edges may repeat [8, p. 22]. Length zero is allowed by Convention 6: the single vertex n_i is a walk of length 0 from n_i to itself.

Example 2.24. In the board graph, n_1, n_2, n_1, n_2, n_3 is a walk of length 4 from n_1 to n_3.

Non-example 2.25. n_1, n_3 is not a walk of the board graph, because n_1n_3 is not an edge. For the textbook's own example, $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ is a walk of its Fig. 1.2.5 and $v_3, v_1, v_4, v_2, v_3, v_5$ is not, because, as the figure shows, v_1v_4 is not an edge there [8, Fig. 1.2.5, p. 21].

In plain words. A round trip ends in the town where it began. Nothing else is asked of it. It may drive a road twice.

Definition 2.26 (closed walk; textbook Definition 1.2.11)**.** A walk $u_0, \ldots, u_k$ is *closed* if $u_0 = u_k$ [8, Def. 1.2.11, p. 22].

Remark 2.27. A closed walk may repeat vertices and edges. A walk of length 0 is closed.

Example 2.28. n_2, n_1, n_2 is a closed walk of the board graph, of length 2.

Non-example 2.29. n_1, n_2, n_3 is a walk that is not closed.

In plain words. A no-repeat-road trip never drives the same road twice. It may still come back to a town it has visited.

Definition 2.30 (trail; the textbook's "path", Definition 1.2.12)**.** A walk is a *trail* if its edges $u_0u_1, \ldots, u_{k-1}u_k$ are pairwise different. The textbook calls this a path [8, Def. 1.2.12, p. 22], and the English word for it is trail.

Remark 2.31. Vertices may repeat in a trail; edges may not.

Example 2.32. The textbook's example $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail of its Fig. 1.2.5 although v_3 occurs twice [8, p. 22].

Non-example 2.33. n_2, n_1, n_2 is not a trail of the board graph, since it uses the edge n_1n_2 twice. The textbook's $v_1, v_3, v_2, v_4, v_3, v_2, v_4, v_3, v_5$ is not a trail either, since it uses v_3v_2 twice [8, p. 22].

In plain words. A no-revisit trip never visits a town twice. Between two towns, such a trip is what the third question asks about.

Definition 2.34 (path; path between n_i and n_j; textbook Definition 1.2.13, simple path)**.** A *path* is a walk whose vertices are pairwise different [8, Def. 1.2.13, p. 22]; the textbook calls it a simple path. A *path between n_i and n_j* is a sequence $u_0, u_1, \ldots, u_k$ of vertices with $u_0 = n_i$, $u_k = n_j$, with $u_{s-1}u_s$ an edge for $1 \leq s \leq k$, and with $u_0, \ldots, u_k$ pairwise different. Its length is k. The two ends n_i and n_j may coincide only when $k = 0$.

Remark 2.35. This is the answer to question (Q3), in the convention fixed by Convention 5. By Convention 6, n_i alone is a path between n_i and n_i, of length 0. Because a path never repeats a vertex, it never repeats an edge, so every path is a trail (Lemma 3.3).

Example 2.36. In the board graph, n_1, n_2, n_3 is a path between n_1 and n_3 of length 2. In the textbook's Fig. 1.2.5, v_1, v_3, v_4, v_5 is a simple path [8, p. 22].

Non-example 2.37. n_1, n_2, n_1, n_2, n_3 is a walk but not a path, because n_1 and n_2 occur twice. The textbook's $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail but not a simple path, because v_3 occurs twice [8, p. 22].

In plain words. A round trip that never drives a road twice may still visit a town twice. A clean round trip does not. It visits no town twice except the start, which is also the end. A clean round trip needs at least three roads.

Definition 2.38 (closed trail, textbook Definition 1.2.14; cycle, textbook Definition 1.2.15)**.** A *closed trail* is a trail that is a closed walk. A *cycle* is a closed trail of positive length in which only the first and last vertices coincide (Convention 6 gives a cycle positive length). The textbook's names are these: its "closed path or cycle" is the closed trail [8, Def. 1.2.14, p. 22], and its "simple cycle" is the cycle [8, Def. 1.2.15, p. 22], which is the English meaning of cycle. Table 2 lists all names.

Remark 2.39. A cycle has at least 3 edges and 3 vertices (Lemma 3.6).

Example 2.40. The textbook's v_1, v_3, v_4, v_2, v_1 is a simple cycle, and $v_3, v_2, v_1, v_3, v_5, v_4, v_3$ is a closed trail that is not simple [8, p. 22].

Non-example 2.41. The board graph has no cycle: it has only two edges, and a cycle needs three (Lemma 3.6). The closed walk n_2, n_1, n_2 is not a closed trail.

In plain words. The distance between two towns is the fewest steps in any trip between them. If no trip joins them, the distance is called infinite.

Definition 2.42 (distance; textbook Definition 1.2.16)**.** The *distance* $d(u, v)$ is the length of a shortest trail from u to v if at least one trail exists, and $+\infty$ otherwise [8, Def. 1.2.16, pp. 22–23].

Remark 2.43. The textbook measures the shortest trail (Definition 2.30). By Lemma 3.5, the shortest walk and the shortest path give the same number, so "shortest path" keeps its usual meaning.

Example 2.44. In the board graph, $d(n_1, n_3) = 2$, given by the path n_1, n_2, n_3.

Non-example 2.45. The number 4 is the length of a walk from n_1 to n_3 but not the distance, because a walk of length 2 exists.

In plain words. A country is one group when you can get from any town to any other town by some trip. If not, the towns split into groups. Within a group every town reaches every other town. Between two groups there is no trip at all.

Definition 2.46 (connected, textbook Definition 2.1.1; connectivity component)**.** G is *connected* if for any two vertices u and v there is a trail from u to v [8, Def. 2.1.1, p. 48]. Write $u \sim v$ if there is a trail from u to v. A relation is *reflexive* if every vertex is related to itself, *symmetric* if $u \sim v$ implies $v \sim u$, and *transitive* if $u \sim v$ and $v \sim w$ imply $u \sim w$. A relation with all three properties is an *equivalence relation*, and the *equivalence class* of a vertex v is the set of all vertices w with $v \sim w$. The textbook notes that $\sim$ has all three properties, without giving a proof [8, p. 49]; reflexivity uses the trail of length 0 (Convention 6), and symmetry is the reversal of a trail (Lemma 3.3). For a set U of vertices, the *induced subgraph* on U keeps the vertices of U and every edge of G with both ends in U. The equivalence classes $V_1, \ldots, V_p$ give the induced subgraphs $G_1, \ldots, G_p$, which the textbook calls the *connectivity components* of G [8, p. 49]. By the textbook's Remark 2.1.1, G is connected if and only if it has exactly one component [8, p. 49].

Remark 2.47. The textbook's word is "path", meaning trail. Lemma 3.4 shows that nothing changes if "trail" is replaced by the course's path, so the board's wording "there exists a path between nodes" defines the same graphs. By Convention 6, the pair $u = v$ is always satisfied by the walk of length 0, so only pairs of different vertices matter.

Example 2.48. The board graph is connected and has one component, $\{n_1, n_2, n_3\}$.

Non-example 2.49. The graph with the single edge n_1n_2 and the isolated vertex n_3 is not connected. Its components are $\{n_1, n_2\}$ and $\{n_3\}$ (Section 6).

3 Question (Q3): paths, and the three words for them

The definition of a path between n_i and n_j is Definition 2.34: a sequence of vertices from n_i to n_j, each two neighbours in the sequence joined by an edge, with no vertex repeated. This section adds only what is new: a comparison of the conventions, and the facts that connect paths to walks. These facts are needed for questions (Q1) and (Q2).

3.1 Three conventions for one family of words

In plain words. Books do not agree on what to call a trip. Some words mean the strict kind, which never visits a town twice. Some mean the looser kind, which never drives a road twice. The table matches the words of the course, of the textbook, and of English.

Table 2: The course, textbook, and English words for walks with different restrictions. The first column says what may not repeat.

What may not repeat	Course word (this note)	Textbook word	English word
nothing	walk	walk, Def. 1.2.10	walk
no edge	trail	path, Def. 1.2.12	trail
no vertex	path	simple path, Def. 1.2.13	path
closed, no edge, first vertex equals last	closed trail	closed path or cycle, Def. 1.2.14	closed trail
closed, only first and last vertex coincide	cycle	simple cycle, Def. 1.2.15	cycle

The textbook rows are Definitions 1.2.10 to 1.2.15 [8, pp. 21–22]. The course column follows Convention 5. The English word for a walk without repeated vertices is path, as in Grinberg's Definition 2.9.1(c) [3, p. 36], and the English cycle is the textbook's simple cycle. The one place where the textbook's wording and the course's wording clash is the word "path": the textbook's path is this note's trail, and the textbook's simple path is this note's path.

3.2 Every walk contains a path

In plain words. Take any trip between two towns. If it visits some town twice, cut out the part between the two visits. What is left is still a trip between the same two towns, and it is shorter. Repeat until no town is visited twice. A trip that never visits a town twice has fewer steps than there are towns.

Lemma 3.1 (a walk contains a path)**.** *Let $w_0, w_1, \dots, w_k$ with $k \geq 0$ be a walk from u to v in a simple graph with N vertices. Then there is a path from u to v of length at most k whose vertices all occur among $w_0, \dots, w_k$. Every path has length at most $N - 1$.*

The textbook states the first part for $u \neq v$ and without a length bound as its Lemma 1.2.1 [8, p. 23]. Grinberg states it with the bound and also for $u = v$ [3, Cor. 2.9.7, p. 39], and states the bound $N - 1$ [3, p. 40]. The proof below is given here.

Proof. We use strong induction on k: to prove the statement for walks of length k we may assume it for walks of every shorter length, not only length $k - 1$. *Base case* $k = 0$. The walk is the single vertex $u = v$, which is a path of length 0.

Induction hypothesis. Let $k \geq 1$ and assume the statement holds for all walks of length less than k.

Step. If $w_0, \dots, w_k$ are pairwise different, the walk is a path of length k and we are done. Otherwise there are positions $p < q$ with $w_p = w_q$. Delete the block $w_{p+1}, \dots, w_q$ and call the result $W' = w_0, \dots, w_p, w_{q+1}, \dots, w_k$. We check that W' is a walk from u to v. Consecutive vertices inside $w_0, \dots, w_p$ and inside $w_{q+1}, \dots, w_k$ are joined as in the original walk, so only the join between w_p and w_{q+1} is new, and there are two cases.

- *Case $q < k$.* Since $w_p = w_q$ and $w_q w_{q+1}$ is an edge, $w_p w_{q+1}$ is an edge. The walk W' ends at $w_k = v$. For example, from the walk n_1, n_2, n_1, n_2, n_3 of the board graph, with $p = 0$ and $q = 2$, the deletion gives n_1, n_2, n_3.
- *Case $q = k$.* There is no w_{q+1}, and $W' = w_0, \dots, w_p$ ends at $w_p = w_q = w_k = v$. For example, from n_1, n_2, n_1 with $p = 0$ and $q = 2 = k$ the deletion gives the single vertex n_1.

In both cases W' starts at $w_0 = u$ (also when $p = 0$, since then W' begins with w_0), ends at v, has length $k - (q - p) < k$, and uses only vertices of the original walk. By the induction hypothesis W' contains a path from u to v of length at most $k - (q - p) < k$ with all its vertices among those of W'. This proves the first part.

For the bound, a path of length k has $k + 1$ pairwise different vertices, and G has only N vertices, so $k + 1 \leq N$, that is $k \leq N - 1$. □

Remark 3.2. If $u \neq v$ and there is a walk of length k from u to v, then there is a path from u to v of length between 1 and $\min(k, N - 1)$. The lemma does not give a path of exactly length k. In the board graph there are two walks of length 4 from n_1 to n_3 and no path of length 4 (Section 6).

3.3 Nesting, reversal, trails and distance

In plain words. A no-revisit trip never drives a road twice, because driving a road twice means visiting its towns twice. So every no-revisit trip is also a no-repeat-road trip. Driving a trip backwards gives a trip of the same number of steps in the other direction.

Lemma 3.3 (nesting and reversal)**.** *(a) Every path is a trail, and every trail is a walk; each inclusion can be strict. (b) If $u_0, \dots, u_k$ is a walk, trail, or path from a to b, then $u_k, \dots, u_0$ is a walk, trail, or path from b to a, of the same length.*

Proof. (a) Suppose a path $u_0, \ldots, u_k$ had two equal edges $u_{s-1}u_s = u_{t-1}u_t$ with $s < t$. The idea is that if an edge appears twice, its two end vertices appear twice, so a vertex is repeated. In detail, since $s < t$, we have $s - 1 < t - 1$ and $s - 1 < t$, so the positions $s - 1$ and $t - 1$ differ, and so do $s - 1$ and t. A path has pairwise different vertices, so u_{s-1} is neither u_{t-1} nor u_t. But the two edges are equal as sets, $\{u_{s-1}, u_s\} = \{u_{t-1}, u_t\}$, so u_{s-1} must equal u_{t-1} or u_t. This is a contradiction. Every trail is a walk by definition. The inclusions are strict: in the board graph n_2, n_1, n_2 is a walk and not a trail, and the textbook's $v_1, v_3, v_2, v_4, v_3, v_5$ is a trail and not a path [8, p. 22]. (b) Edges are unordered pairs, and the conditions on repeated edges and vertices do not depend on the order of the list. □

In plain words. Every no-revisit trip is also a no-repeat-road trip. Now go the other way. If some no-repeat-road trip joins two towns, then some no-revisit trip joins them too. So two towns can reach each other by one kind of trip exactly when they can by the other.

Lemma 3.4 (trail if and only if path)**.** *For vertices u and v there is a trail from u to v if and only if there is a path from u to v if and only if there is a walk from u to v. Hence Definition 2.46 defines the same connected graphs whether "trail" is replaced by "path" or by "walk".*

Proof. A path is a trail and a trail is a walk (Lemma 3.3). A walk from u to v contains a path from u to v (Lemma 3.1), which closes the circle of implications. □

In plain words. The fewest steps of any trip between two towns is the same number whichever kind of trip we allow. So the word "distance" means one thing.

Lemma 3.5 (distance)**.** *For vertices u and v, let W, T, and P be the least lengths of a walk, a trail, and a path from u to v, each taken as $+\infty$ if none exists. Then $W = T = P$. In particular the textbook's distance (Definition 2.42) is T, and equals the length of a shortest walk and of a shortest path.*

Proof. Every path is a trail and every trail is a walk, so $W \leq T \leq P$. If W is finite, a walk of length W contains a path of length at most W (Lemma 3.1), so $P \leq W$. Hence $W = T = P$. If $W = +\infty$, there is no walk and so no trail and no path, and all three are $+\infty$. □

Grinberg makes the same remark for walks and paths [3, p. 40].

In plain words. A clean round trip needs at least three roads. One road is not enough, because a road cannot start and end in the same town. Two roads are not enough, because the trip would drive one road there and back.

Lemma 3.6 (short cycles)**.** *In a simple graph a closed trail of positive length has at least* 3 *edges. Hence a cycle has at least* 3 *edges and* 3 *different vertices.*

Proof. A closed trail $u_0, \ldots, u_k$ with $k \geq 1$ cannot have $k = 1$, since $u_1 = u_0$ would make u_0u_1 a loop. It cannot have $k = 2$, since u_0, u_1, u_0 uses the edge u_0u_1 twice. So $k \geq 3$. In a cycle only the first and last vertex coincide, so $u_0, \ldots, u_{k-1}$ are $k \geq 3$ different vertices. □

The textbook writes C_n with $n \geq 3$ for the simple cycle on n vertices [8, p. 22].

4 Question (Q1): what the entry of a power of A represents

The answer to question (Q1), with the mark above A read as an exponent k (Convention 3), is that $(A^k)_{ij}$ is a count of walks. Walks are defined in Definition 2.22, and the product that forms A^k in Definition 2.13.

4.1 The walk-counting theorem

In plain words. Ask how many trips of exactly k steps lead from one town to another. The road table answers for one step: the cell holds 1 if there is a road and 0 if not. The table multiplied by itself answers for two steps, and so on. A trip may turn back and may revisit towns, and every such trip is counted.

Theorem 4.1 (walk counting)**.** *Let G be a simple graph with adjacency matrix A. For every integer $k \geq 1$ and all i, j (including $i = j$), the entry $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j.*

Theorem 1.1.3 of the University of Galway notes [6, p. 4], Proposition 2 of Gotti's lecture notes [2, p. 1], and Theorem 1 of Bai's lecture notes [1, p. 1] state the same result. The textbook has no such theorem (Convention 8). The proof below is given here.

Proof. We use induction on k.

Base case $k = 1$. A walk of length 1 from n_i to n_j is a single edge $n_i n_j$. In a simple graph there is at most one such edge, so the number of walks of length 1 is 1 if n_i and n_j are adjacent and 0 otherwise. This is $A_{ij} = (A^1)_{ij}$ by Definition 2.9.

Induction hypothesis. Assume for some $k \geq 1$ that, for all i and j, $(A^k)_{ij}$ is the number of walks of length k from n_i to n_j.

Step. By Definition 2.13, $A^{k+1} = A^k A$, so

$$(A^{k+1})_{ij} = \sum_{r=1}^{N} (A^k)_{ir} A_{rj}. \tag{1}$$

Sort the walks of length $k+1$ from n_i to n_j by their last-but-one vertex n_r. Such a walk consists of a walk of length k from n_i to n_r, followed by one step from n_r to n_j, and cutting off the last edge recovers the two parts uniquely. The final step is possible in exactly A_{rj} ways, which is 1 if $n_r n_j$ is an edge and 0 otherwise. By the induction hypothesis the first part can be chosen in $(A^k)_{ir}$ ways. This is a one-to-one correspondence between the walks $w_0, \ldots, w_k, w_{k+1}$ with $w_k = n_r$ and the pairs (walk $w_0, \ldots, w_k$ of length k from n_i to n_r, final step w_k, w_{k+1} with $w_{k+1} = n_j$). Every such pair gives a walk of length $k+1$, because the final step needs only that $n_r n_j$ is an edge and puts no condition on the first part, and the two choices are independent. So for fixed r the number of such walks is $(A^k)_{ir} A_{rj}$. Different values of r give different walks, since the last-but-one vertex differs, and every walk has some last-but-one vertex. Adding over r gives the right side of (1). So $(A^{k+1})_{ij}$ is the number of walks of length $k+1$ from n_i to n_j, which completes the induction. □

Remark 4.2. With Convention 6 the statement also holds for $k = 0$: $A^0 = I$, there is one walk of length 0 from n_i to itself and none from n_i to a different vertex. So the base case $k = 0$ holds because $A^0 = I$, and the step of the proof then applies from $k = 0$ unchanged. This is the note's own convention. The textbook's definitions of walks do not say whether length 0 is allowed, but its usage assumes it (Convention 6).

Remark 4.3. An entry of A^k is a count, not a walk. A positive entry $(A^k)_{ij}$ means that a walk of length k from n_i to n_j exists, and so, by Lemma 3.1, that a path from n_i to n_j of length at most k exists. It does not mean that a path of length exactly k exists (Remark 3.2). The matrix A^k is symmetric for every k, because reversing a walk from n_i to n_j gives a walk from n_j to n_i of the same length (Lemma 3.3(b)), so the two counts are equal.

4.2 Walks, not paths

In plain words. The product adds up, over every middle town, the ways to get there and the ways to go on. It never asks whether a town was visited before. So a trip that goes out and straight back is counted. The cell for a town that has at least one road, and itself, is therefore not zero, though no no-revisit trip of two steps can start and end at one town.

Theorem 4.4 (the diagonal of A^2)**.** *For every i, $(A^2)_{ii} = \deg(n_i)$. If $\deg(n_i) \geq 1$ then $(A^2)_{ii} \geq 1$, although no path of length 2 joins n_i to n_i. So $(A^2)_{ii}$ counts walks and not paths.*

Galway's notes state that the number of walks of length 2 from a vertex to itself equals the degree of the vertex [6, p. 4]. The proof is given here.

Proof. By Definition 2.13, $(A^2)_{ii} = \sum_r A_{ir}A_{ri}$. Since $A_{ri} = A_{ir}$ (Definition 2.9) and every entry is 0 or 1, each term equals $A_{ir}^2 = A_{ir}$. So $(A^2)_{ii} = \sum_r A_{ir} = \deg(n_i)$ by Theorem 2.21. A path of length 2 has 3 pairwise different vertices, but its first and last vertices would both be n_i. So there is no such path. The only path from n_i to n_i is the one of length 0 (Convention 6), which is not of length 2. □

Remark 4.5. The same happens off the diagonal. In the board graph, $(A^3)_{12} = 2$, counted by the walks n_1, n_2, n_1, n_2 and n_1, n_2, n_3, n_2. Both repeat a vertex and an edge, and no path has length 3, because a path in a graph with 3 vertices has length at most 2 (Lemma 3.1). The matrices are in Section 6. Counting paths is not what a matrix power does.

5 Question (Q2): deciding connectivity from the adjacency matrix

The board defines a connected graph by the existence of a path between any two vertices [4], and the textbook gives the same definition in terms of its own word "path", which is a trail (Definition 2.46). Lemma 3.4 shows that the choice of word does not change which graphs are connected. This section turns the definition into a test that reads only A.

Notation. Let $S = A + A^2 + \cdots + A^{N-1}$. For $N = 1$ the sum is empty and S is the zero matrix. The matrix I is the identity matrix of Definition 2.13.

5.1 The sum test

In plain words. To know whether every town can reach every other town, add up the tables for one step, two steps, and so on up to $N - 1$ steps. Look at the cells for two different towns, not at a town and itself. If none of these cells is zero, every town can reach every other. If one of them is zero, that pair of towns cannot reach each other. Why stop at $N - 1$ steps? A trip that never visits a town twice has fewer steps than there are towns.

Theorem 5.1 (sum test)**.** *A simple graph with N vertices and adjacency matrix A is connected if and only if $S_{ij} > 0$ for all $i \neq j$.*

Paramadevan and Sotheeswaran state this result as their Theorem 2 [7, p. 17]. Their proof counts "different paths" and does not use the step from a walk to a path, so it is not used here; the proof below is given here and rests on Theorem 4.1, Lemma 3.1 and Lemma 3.4.

Proof. If $N = 1$ there is no pair with $i \neq j$, so the condition holds; and a graph with one vertex is connected, since its only pair is $u = v$ (Convention 6). Let $N \geq 2$ and fix $i \neq j$. By Theorem 4.1 every entry of every A^k is a count, so a nonnegative integer. So S_{ij} is a sum of nonnegative integers, and $S_{ij} > 0$ if and only if $(A^k)_{ij} \geq 1$ for some k with $1 \leq k \leq N - 1$.

Claim: $S_{ij} > 0$ if and only if there is a walk from n_i to n_j. Suppose $S_{ij} > 0$. Choose k with $1 \le k \le N-1$ and $(A^k)_{ij} \ge 1$. By Theorem 4.1 there are $(A^k)_{ij} \ge 1$ walks of length k from n_i to n_j, so a walk exists. Conversely, suppose a walk exists. By Lemma 3.1 there is a path from n_i to n_j, of some length k with $k \le N-1$. Since $i \ne j$ the path has at least two vertices, so $k \ge 1$. A path is a walk, so $(A^k)_{ij} \ge 1$ by Theorem 4.1. All other terms of S_{ij} are nonnegative, so $S_{ij} \ge 1$. This proves the claim.

Now suppose G is connected. For $i \ne j$ there is a trail from n_i to n_j (Definition 2.46), hence a walk, hence $S_{ij} > 0$ by the claim. Conversely, suppose $S_{ij} > 0$ for all $i \ne j$. By the claim there is a walk, and so by Lemma 3.4 a trail, between any two different vertices. Pairs with $u = v$ are covered by the walk of length 0 (Convention 6). So G is connected. □

In plain words. The zeros of the sum tell you which pairs of towns cannot reach each other. Pairs that can reach each other are in the same group of towns.

Lemma 5.2 (components from S)**.** *For $i \ne j$, $S_{ij} > 0$ if and only if n_i and n_j lie in the same connectivity component. So the zero entries of S off the diagonal mark exactly the pairs of vertices in different components, and G is connected if and only if there is no such zero.*

Proof. By the claim in the proof of Theorem 5.1, $S_{ij} > 0$ if and only if a walk from n_i to n_j exists. By Lemma 3.4 this holds if and only if a trail exists, that is, $n_i \sim n_j$ (Definition 2.46), which means that n_i and n_j lie in one equivalence class. □

5.2 The same test with one matrix power

In plain words. There is a shortcut. Add a 1 to every town's own cell in the road table, then multiply together $N-1$ copies of that new table. The country is one group exactly when no cell of the result is zero. The extra 1s let trips of fewer steps count too.

Theorem 5.3 (power test)**.** *A simple graph with N vertices is connected if and only if every entry of $(I+A)^{N-1}$ is positive.*

Gotti's Proposition 3 [2, p. 2] and Bai's Theorem 2 [1, p. 1] state this result. Gotti's proof rests on the bound $N-1$ on the length of a path and on the binomial identity [2, p. 2]; the proof below is given here and also covers the step from a walk to a path.

Proof. The binomial theorem $(X+Y)^m = \sum_{k=0}^{m} \binom{m}{k} X^k Y^{m-k}$ holds for two matrices that commute, $XY = YX$, where $\binom{m}{k}$ is the number of ways to choose k of the m factors. The matrices I and A commute, since $IA = A = AI$, so with $X = A$, $Y = I$, and $m = N-1$ we may expand $(I+A)^{N-1}$ as a weighted sum of $A^0, \dots, A^{N-1}$. This gives every walk length up to $N-1$ a positive weight:

$$(I+A)^{N-1} = \sum_{k=0}^{N-1} \binom{N-1}{k} A^k, \qquad A^0 = I. \tag{2}$$

Every coefficient $\binom{N-1}{k}$ is an integer and is at least 1 for $0 \le k \le N-1$, and every entry of every A^k is a nonnegative integer. In (2), on the diagonal the term $k = 0$ contributes $\binom{N-1}{0} I_{ii} = 1$, so every diagonal entry is at least 1 for every graph. Off the diagonal, $I_{ij} = 0$, so the entry equals $\sum_{k=1}^{N-1} \binom{N-1}{k} (A^k)_{ij}$. If $S_{ij} > 0$, some $(A^k)_{ij} \ge 1$ with $1 \le k \le N-1$, and its coefficient is at least 1, so the entry is positive. If the entry is positive, some $(A^k)_{ij} \ge 1$ with $1 \le k \le N-1$, so $S_{ij} > 0$. So all entries of $(I+A)^{N-1}$ are positive if and only if $S_{ij} > 0$ for all $i \ne j$, which by Theorem 5.1 holds if and only if G is connected. □

Remark 5.4. The entries of $(I+A)^{N-1}$ and of S are sums of walk counts, in the case of $(I+A)^{N-1}$ with binomial weights. They are not counts of paths. Only their positivity is used by the tests. Section 7 checks two published sentences that read them as path counts.

5.3 What the test does not do

In plain words. One step is not enough, and neither is one fixed number of steps. Some pairs of towns need one step, others need two. That is why the tables for every number of steps are added. Also, the cell of a town with itself says nothing about the groups.

Remark 5.5 (one power is not a test). In the board graph, $N-1=2$ and $(A^2)_{12}=0$ although n_1n_2 is an edge. The reason in words: every edge joins n_2 to one of n_1 and n_3, so every walk from n_1 to n_2 takes an odd number of steps, and A^2 counts only walks of two steps. On the other hand $(A^1)_{13}=0$ while $(A^2)_{13}=1$, so the pair n_1, n_3 needs the second power and the pair n_1, n_2 needs the first. Both A and A^2 are needed. The limit $N-1$ cannot be lowered in general: in a graph with vertices $n_1, \ldots, n_N$ and edges n_sn_{s+1} for $1 \le s < N$, each step changes the index by 1, so every walk from n_1 to n_N has at least $N-1$ steps. For $N=4$, the vertices n_1, n_2, n_3, n_4 lie in a row. The only walk of three steps from n_1 to n_4 is n_1, n_2, n_3, n_4, and no walk of one or two steps joins them, so $(A^1)_{14}=(A^2)_{14}=0$ and $(A^3)_{14}=1$, as the script confirms [10].

Remark 5.6 (reading "any two vertices" without Convention 6). Without Convention 6, a trail of length 0 would not exist, and a vertex that lies on no closed trail of positive length has no trail of positive length to itself. Then "any two vertices" in Definition 2.46 would have to be read as "any two different vertices", since otherwise every graph with such a vertex, including the board graph, would fail the definition for the pair $u=v$ alone. Under that reading the definition and the tests of this section agree, because the proof of Theorem 5.1 uses only pairs of different vertices, and for $N=1$ there are no such pairs, so the graph with one vertex is connected under either reading.

Remark 5.7 (the diagonal says nothing). By Convention 6 every vertex reaches itself by a walk of length 0, so a pair $u=v$ carries no information about connectivity, and neither does the diagonal of S. The connected graph with two vertices and one edge has $S=\begin{pmatrix}0&1\\1&0\end{pmatrix}$, with zero diagonal. The disconnected graph of Section 6 has diagonal $(1,1,0)$, and the connected board graph has diagonal $(1,2,1)$. This is why Theorem 5.1 reads only entries with $i \ne j$, and why $(I+A)^{N-1}$, whose diagonal entries are always positive, can be read as "no zero anywhere".

5.4 An alternative: breadth-first search

This subsection is for comparison only; the answers to the three questions do not need it.

In plain words. There is another way to find out. Stand in one town and mark it. Then mark every town joined to a marked town, again and again, until no new town can be marked. If every town is marked, the country is one group.

Breadth-first search (BFS) starts at n_1, marks it, and keeps a waiting line of marked vertices; it repeatedly takes the first vertex of the line and marks and queues each of its unmarked neighbours. The CS 341 lecture slides of the University of Waterloo state that, for every vertex v, there is a path from the start to v if and only if v is marked at the end [9, slide 11], and that with adjacency lists the running time is proportional to the number of vertices plus the number of edges [9, slide 8]. The slides list connectivity as an application without proof, so this note does not claim a proof from them. The following step is derived here, using the slides' statement about marked vertices: G is connected if and only if all N vertices are marked. If G is connected, every vertex is joined to n_1 by a path, so every vertex is marked. If all vertices are marked, n_1 reaches every vertex, so two vertices u and v are joined through n_1, by a walk from u to n_1 (the reverse of a walk from n_1 to u, Lemma 3.3) followed by a walk from n_1 to v, and a walk gives a path by Lemma 3.1. When the input is the matrix A, each vertex is taken from the line at most once, because it is queued only when it is marked for the first time, so there are at most N such vertices; the search scans one row of N entries for each, which is at most $N \cdot N = N^2$ entries in total (derived). BFS needs no matrix powers and also lists the component

of n_1. Its purpose here is to give a second route to question (Q2), independent of matrix powers, against which the matrix tests can be compared; the question on the board asks for a test from A by algebra.

6 Worked examples and the board mismatch

Every matrix below was computed by the script described in Appendix A, and every entry that is interpreted was matched against an explicit list of walks.

6.1 Example 1, connected: the board graph

In plain words. Three towns stand in a row. The middle town has a road to each end town. The two end towns have no road between them. You can get from any town to any other, but the ends meet only through the middle.

The board graph has vertices n_1, n_2, n_3 and edges n_1n_2 and n_2n_3 (Figure 1). Its adjacency matrix is the matrix of Example 2.11. The powers are

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad A^3 = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 2 \\ 0 & 2 & 0 \end{pmatrix}.$$

For $N = 3$ the sum test uses $S = A + A^2$, and the power test uses $(I + A)^2$:

$$S = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \qquad (I + A)^2 = \begin{pmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{pmatrix}.$$

Reading the entries. By Theorem 4.1, each entry counts walks.

- $(A^2)_{13} = 1$ counts the single walk n_1, n_2, n_3. Note that $A_{13} = 0$: there is no edge n_1n_3, yet the second power is positive, because a walk of two steps exists.
- $(A^2)_{11} = 1$ counts the walk n_1, n_2, n_1. $(A^2)_{22} = 2$ counts the walks n_2, n_1, n_2 and n_2, n_3, n_2. $(A^2)_{33} = 1$ counts n_3, n_2, n_3. By hand, $(A^2)_{22} = A_{21}A_{12} + A_{22}A_{22} + A_{23}A_{32} = 1+0+1 = 2$. The diagonal $(1, 2, 1)$ equals the degrees $\deg(n_1), \deg(n_2), \deg(n_3)$, as Theorem 4.4 says. These are walks and not paths: none of them is a path, and none is even a trail, since each uses an edge twice.
- $(A^3)_{12} = 2$ counts the walks n_1, n_2, n_1, n_2 and n_1, n_2, n_3, n_2 (Remark 4.5). There is no path of length 3 from n_1 to n_2. Also $(A^2)_{12} = 0$ although n_1n_2 is an edge, for the parity reason of Remark 5.5.
- $(A^4)_{13} = 2$ counts the walks n_1, n_2, n_1, n_2, n_3 and n_1, n_2, n_3, n_2, n_3. No path of length 4 joins n_1 and n_3, although a path of length 2 does (Remark 3.2).

The tests. The six entries of S with $i \neq j$ are all 1, so by Theorem 5.1 the graph is connected. For example, $S_{13} = A_{13} + (A^2)_{13} = 0 + 1 = 1$. All nine entries of $(I + A)^2$ are positive, so Theorem 5.3 gives the same answer. The entries of S and of $(I + A)^2$ are not path counts: $S_{22} = 2$ and $((I + A)^2)_{11} = 2$, but at most one path (the one of length 0) joins a vertex to itself.

6.2 Example 2, disconnected: one edge and an isolated vertex

In plain words. Two towns have a road between them. A third town has no road at all. Nobody can reach the third town, and it cannot reach anyone.

Take vertices n_1, n_2, n_3 and the single edge $n_1 n_2$. Then

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (I + A)^2 = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The entries S_{13}, S_{23}, S_{31}, and S_{32} are 0, so by Theorem 5.1 the graph is not connected. By Lemma 5.2 these zeros mark exactly the pairs in different components, and the components are $\{n_1, n_2\}$ and $\{n_3\}$. The same four positions are zero in $(I + A)^2$, so Theorem 5.3 agrees. Breadth-first search from n_1 marks n_1 and n_2 only, which agrees too. The diagonal of S has a zero at position $(3, 3)$. This says nothing about n_3 reaching itself, which illustrates Remark 5.7.

6.3 The board drawing and the board matrix

In plain words. The picture on the board and the table on the board do not show the same country. The picture shows three roads. The table shows two. This note follows the table, and says so. It also shows that the answer to the question about one group is the same either way.

The drawing on the board shows three circled vertices labelled 1, 2, 3 and three lines [4, 5]. The two photographs show the same board and were read together. Figure 2 redraws it as read from the photographs, and Table 3 compares the drawing and the matrix edge by edge.

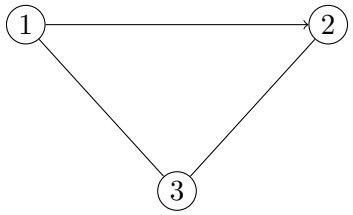


Figure 2: The board drawing as read from the photographs: three vertices, a line from 1 to 2 with an arrowhead at 2, a line from 1 to 3, and a line from 2 to 3. This is a redrawing by the author of this note, not a copy.

Table 3: The board drawing compared with the board matrix, edge by edge. The matrix has $A_{12} = A_{21} = A_{23} = A_{32} = 1$ and all other entries 0.

Pair of vertices	Drawing	Matrix
1 and 2	a line, with an arrowhead at 2	edge ($A_{12} = A_{21} = 1$)
1 and 3	a line	no edge ($A_{13} = A_{31} = 0$)
2 and 3	a line	edge ($A_{23} = A_{32} = 1$)

The one disagreement is the pair 1 and 3. The arrowhead is read as a stray mark (Convention 4): the matrix has $A_{12} = A_{21} = 1$, so the connection runs both ways, and the problem concerns undirected graphs. The note takes the matrix as the example, because the board says the graph is given through its adjacency matrix (Convention 4). If the drawing were taken instead, the graph would be the triangle with all three edges. For the triangle, the script gives

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad S = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \quad (I + A)^2 = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}.$$

Both tests say connected, so the answer to question (Q2) for the board's example does not depend on how the mismatch is read. Which reading the lecturer meant cannot be settled from the photographs.

7 Discussion: limits, checks, and published sentences that fail

Scope. The results are stated for finite simple undirected graphs, and directed graphs and multigraphs are not covered by this note.

What rests on conventions. Three choices are the note's own or are interpretations, and each is listed in Section 1. The mark above A is read as an exponent k (Convention 3). The matrix is taken as the example and not the drawing (Convention 4). A walk of length 0 is allowed (Convention 6); the textbook's definitions of walks do not say, but its statements on distance and on the relation $\sim$ assume it. The tests of Theorem 5.1 and Theorem 5.3 read only pairs of different vertices, and for such pairs a walk has positive length. If length 0 were not allowed, the definition of connected would have to read "any two different vertices", and the tests would still agree with it (Remark 5.6). So the answer to question (Q2) does not depend on Convention 6. What depends on it are the case $k = 0$ of Theorem 4.1, the reflexivity of $\sim$ and hence the description of components as its classes, the statement of Lemma 3.1 for $u = v$, and the remark on the diagonal.

State of the sources. The textbook has no walk-counting theorem and no matrix test; this is the result of a text search, not a proof (Convention 8). Of the sources consulted, only Paramadevan and Sotheeswaran state the sum form of the test, in a regional journal [7, p. 17], and their proof is not used here. Gotti and Bai state the power form [2, 1]. The proofs of Theorem 5.1 and Theorem 5.3 therefore rest on the arguments of this note. Support for breadth-first search is lecture slides only [9].

Experimental check. A script compared the sum test, the power test, and breadth-first search on every labelled simple graph with $N = 1$ to 6 vertices, which is 1, 2, 8, 64, 1024, and 32768 graphs, of which 1, 1, 4, 38, 728, and 26704 are connected. The three methods agreed on every graph (Appendix A) [10]; an earlier script gives the same counts [11]. The same script also compared every entry of A^k for $k = 0$ to 4 with an explicit enumeration of walks for all graphs with 1 to 4 vertices: 5525 entries and no failures [10]. The number 5525 is $5 \cdot 1105$: there are 5 values of k, and the graphs with 1, 2, 3, 4 vertices number 1, 2, 8, 64 with 1, 4, 9, 16 matrix entries each, so $1 \cdot 1 + 2 \cdot 4 + 8 \cdot 9 + 64 \cdot 16 = 1105$. These are experimental findings for small N, a check of the statements and not a proof of them. The proofs are in Sections 4 and 5.

Three published sentences that do not hold under the standard meaning of "path". Three sentences in the sources read an entry of a matrix power, or of a sum of powers, as a count of paths. Each is quoted, restated in this note's notation, and checked against a counterexample on the board graph, with the matrices of Section 6. The counterexamples use "path" in the standard sense, no repeated vertex, as in Definition 2.34 and in Grinberg's Definition 2.9.1(c) [3, p. 36]. The quotations are the words of the sources, with the symbols typeset in math mode. None of the sentences is used in the proofs above. The tests that sentences 2 and 3 accompany are correct, as proved above, because only the positivity of entries enters.

1. Paramadevan and Sotheeswaran, Theorem 1, p. 16 [7]: "the number of different paths of length l between the ith and jth vertices of G is equal to the (i, j)th entry of A_G^l." The paper does not define "path", and in its proof for $l = 2$ it identifies the (i, i) entry of A^2, which it says equals the degree of the ith vertex, with a number of paths of length

2 between the ith vertex and itself. So the sentence is read here in two ways. If "path" means a walk, as that proof uses it, the sentence is Theorem 4.1, and the paper's own property 7 on p. 16, that the (i, j)th entry of A^m "is equal to the number of walks of length m from v_i to v_j", says the same. If "path" has its standard meaning, the restated sentence, "$(A^l)_{ij}$ is the number of paths of length l from n_i to n_j", is false. In the board graph, $(A^2)_{11} = 1$, but no path of length 2 joins n_1 to n_1 (Theorem 4.4); and, off the diagonal, $(A^3)_{12} = 2$, but no path of length 3 joins n_1 to n_2, since a path in a graph with 3 vertices has length at most 2 (Remark 4.5). The second counterexample does not depend on Convention 6.

2. Paramadevan and Sotheeswaran, proof of Theorem 2, p. 17 [7]: "b_{ij} = number of different paths of length less than n from the vertex i to j." Under the walk reading, with "length less than n" meaning lengths 1 to $n - 1$ as in the sum the source writes out just before, this holds for $B = A + \cdots + A^{N-1}$. Under the standard reading, restated as "B_{ij} is the number of paths of length less than N from n_i to n_j", it fails. In the board graph $N = 3$ and $B = A + A^2$ has $B_{22} = 2$, but at most one path of length less than 3 joins n_2 to itself, namely the path of length 0 (Convention 6). If length 0 is not allowed there is none, so the sentence fails under either reading of length 0.

3. Bai, proof sketch of Theorem 2, p. 2 [1]: "$(I+A)^{n-1}$ aggregates all paths of length $\leq n-1$." Bai's Theorem 1, p. 1, itself says that the entries of A^k count walks and not paths [1], so "paths" is used there in the standard sense. Restated as "the entry R_{ij} of $R = (I + A)^{N-1}$ is the number of paths of length at most $N - 1$ from n_i to n_j", the sentence fails: in the board graph $R = (I + A)^2$ has $R_{11} = 2$, but only one path of length at most 2 joins n_1 to itself, the path of length 0.

8 Conclusion

The three questions have the following answers, for finite simple undirected graphs with the conventions of Section 1.

(Q1) The entry $(A^k)_{ij}$ of the k-th power of the adjacency matrix is the number of walks of length exactly k from n_i to n_j (Theorem 4.1). It counts walks, in which vertices and edges may repeat, and not paths: $(A^2)_{ii}$ equals the degree of n_i, which is positive for every vertex with a neighbour, although no path of length 2 joins a vertex to itself (Theorem 4.4). The proof is by induction on k, grouping walks by their last-but-one vertex.

(Q2) A graph with N vertices is connected if and only if every entry off the diagonal of $A + A^2 + \cdots + A^{N-1}$ is positive (Theorem 5.1), and equivalently if and only if $(I + A)^{N-1}$ has no zero entry (Theorem 5.3). The proof combines walk counting with the fact that every walk contains a path and that a path has at most $N - 1$ edges (Lemma 3.1). The zero entries of the sum mark exactly the pairs of vertices in different components (Lemma 5.2). On the board's matrix the test says connected, and on the graph with one edge and an isolated vertex it says not connected (Section 6). Breadth-first search gives the same answer by another route.

(Q3) In the course, a path between n_i and n_j is a sequence of vertices from n_i to n_j, each two neighbours joined by an edge, with no vertex repeated (Definition 2.34); it is the textbook's simple path, Definition 1.2.13. The textbook's own word "path" (Definition 1.2.12) means a walk with no repeated edge, called a trail here. For deciding connectivity the three kinds of route, walk, trail, and path, join exactly the same pairs of vertices (Lemma 3.4).

Two points were reported and not settled. The board's drawing has three lines and its matrix has two edges, and the note follows the matrix (Table 3); both readings give a connected graph.

The textbook's definitions do not say whether a walk of length 0 exists, though its statements on pp. 23 and 49 assume one; the note adopts it as its own convention; no answer to question (Q2) depends on that choice.

References

[1] Shuliang Bai. *Important Properties and Applications of the Adjacency Matrix (20250310).* BIMSA lecture notes, dated 2025-03-10. 2025. URL: `https://bimsa.net/doc/notes/48535.pdf` (visited on Nov. 6, 2026).

[2] Felix Gotti. *MIT 18.211 Combinatorial Analysis, Lecture 27: Adjacency Matrices and The Matrix-Tree Theorem.* MIT lecture notes. No usable date. n.d. URL: `https://math.mit.edu/~fgotti/docs/Courses/C.%20Combinatorial%20Analysis/27.%20The%20Matrix-Tree%20Theorem/The%20Matrix-Tree%20Theorem.pdf` (visited on Nov. 6, 2026).

[3] Darij Grinberg. *An introduction to graph theory (Text for Math 530, Spring 2022, Drexel University).* arXiv:2308.04512v3, Spring 2025 edition, version June 7, 2025. Preprint. 2025. URL: `https://arxiv.org/pdf/2308.04512` (visited on Nov. 6, 2026).

[4] *Lecture whiteboard photo 1.* Photograph, file `sources/whiteboard-1.jpg`. Kept with the paper as source material. Author and date unknown. n.d.

[5] *Lecture whiteboard photo 2.* Photograph, file `sources/whiteboard-2.jpg`. Same board as photo 1, different angle. Kept with the paper as source material. Author and date unknown. n.d.

[6] *Linear algebra notes, section 1.1 The Adjacency Matrix (University of Galway).* University of Galway mathematics lecture notes, Chapter 1, file section1-1.pdf. No author or date shown in the file. n.d. URL: `https://maths.nuigalway.ie/~rquinlan/linearalgebra/section1-1.pdf` (visited on Nov. 6, 2026).

[7] P. Paramadevan and S. Sotheeswaran. "Properties of Adjacency Matrix of a Graph and It's Construction". In: *Journal of Science, Eastern University, Sri Lanka (JSc EUSL)* 12.1 (2021), pp. 13–21. DOI: `10.4038/jsc.v12i1.29`. URL: `https://www.fsc.esn.ac.lk/jsc/archive/12.1.2.pdf` (visited on Nov. 6, 2026).

[8] P. A. Petrosyan, V. V. Mkrtchyan, and R. R. Kamalyan. *Graph Theory, study manual (in Armenian).* Course textbook, file `Grafneri_tesutyun.pdf`. Cited by printed page number, which is the PDF page number minus one. Yerevan: Yerevan State University Press, 2015. ISBN: 978-5-8084-1995-7.

[9] Eric Schost. *CS 341: Algorithms, Module 3: breadth-first search, depth-first search (slides).* University of Waterloo lecture slides. Year not shown. n.d. URL: `https://cs.uwaterloo.ca/~eschost/publications/CS341-lec03-Eric.pdf` (visited on Nov. 6, 2026).

[10] *Verification script for every matrix, sum, count and exhaustive check printed in the paper.* Supplementary script `appendices/compute.py` with output `appendices/compute.out`. Supplementary verification script written by the author of this paper. Run with python3 -I. 2026.

[11] *Verification script for the matrices and the exhaustive connectivity check.* Supplementary script `modules/m3_check.py` with output `modules/m3_check.out`. Supplementary verification script written for this paper. Run with python3 -I. n.d.

A Sources, pages checked, and how to rerun the calculations

Sources. The course textbook, an Armenian study manual by Petrosyan, Mkrtchyan and Kamalyan [8], has its Armenian title omitted from the bibliography because the typesetting

here does not support the script, and the file is `Grafneri_tesutyun.pdf`. The textbook pages viewed as images for the definitions quoted are the printed pages 11, 16, 21, 22, 23, 48, and 49. The text search described in Convention 8 hit the printed pages 7, 11, 12, 34, and 61; of these only page 11 was viewed as an image for this note, and the text layer of the others contains neither result. The lecture board is the pair of photographs [4, 5], both viewed. The other sources are lecture notes and a journal article, listed in the bibliography with the date they were accessed, 2026-11-06 [6, 2, 3, 1, 7, 9]. Grinberg's text is a preprint and is used for statements only; its proofs are not used.

Calculations. Every matrix, sum, and count in the text is printed by the script `appendices/compute.py` (pure Python, no packages), whose output is saved in `appendices/compute.out`. To rerun it, change to the paper folder and execute

```
python3 -I appendices/compute.py
```

The run takes about twenty seconds, most of it in the exhaustive check over all graphs with up to 6 vertices. An earlier script, `modules/m3_check.py`, computed the same matrices for the two examples and the exhaustive check, and its output agrees [11]. The script `appendices/compute.py` [10] also lists, for the board graph, the walks, trails, and paths of given length between given vertices, which are the lists quoted in Section 6.

Key output of the script. The last two blocks of `appendices/compute.out`, which support the experimental statements in Section 7, read as follows.

```
=== Exhaustive check over all labelled simple graphs, N = 1..6
N=1: graphs 1, connected 1, disagreements 0
N=2: graphs 2, connected 1, disagreements 0
N=3: graphs 8, connected 4, disagreements 0
N=4: graphs 64, connected 38, disagreements 0
N=5: graphs 1024, connected 728, disagreements 0
N=6: graphs 32768, connected 26704, disagreements 0

=== Walk-count theorem against enumeration, all graphs N = 1..4, k = 0..4
entries compared: 5525  failures: 0
```

Here "disagreements" counts graphs on which the sum test, the power test, and breadth-first search do not all give the same answer. The script also prints the four-vertex path powers $(A^1)_{14} = 0$, $(A^2)_{14} = 0$, $(A^3)_{14} = 1$ used in Remark 5.5.

Building the paper. The document is compiled with `latexmk -pdf` and `biber`, which come with TeX Live (see `README.md`).